

The TARGET guideline for reporting observational studies of interventions



Observational (non-experimental) studies of interventions can generate valuable evidence when randomized trials are infeasible, unethical or unable to provide timely answers to important clinical and policy questions. The use of observational data to estimate the comparative benefits and harms of medical interventions has increased substantially and has informed clinical, policy and regulatory decisions^{1,2}. This growth has been driven in part by the expanding availability of large healthcare databases and methodological advancements^{1,3}. However, causal inferences made from observational data are often criticized as being at high risk of bias.

The practice of designing observational analyses to mimic a randomized trial dates back to the 1950s and has long been used to compare the effectiveness of medical interventions⁴. The approach was generalized to time-varying treatments in 1986 (ref. 5) and was subsequently formalized in 2016 within the ‘target trial’ framework³. The target trial framework helps researchers to explicitly specify all the protocol components of a hypothetical pragmatic randomized trial – the target trial – that would answer the causal question of interest and can be emulated closely through the use of observational data³. The protocol of the target trial is then mapped to, or ‘emulated’ with, the observational data³. Acknowledging the benefits of the target trial framework, the UK National Institute for Health and Care Excellence⁶ and Canada’s Drug Agency⁷ both encourage use of the framework when observational data are presented as evidence of the effects of medical interventions.

Explicitly specifying and emulating a target trial reduces common yet often avoidable design-related flaws in observational analyses that may lead to selection and time-related bias⁸. These biases can occur when an observational study departs from how a randomized trial would normally be designed – for example, when people are assigned to treatment groups on the basis of information that becomes available after the start of follow-up. Such inadvertently investigator-introduced

biases can explain differences between effect estimates from observational data and those from randomized trials⁹. For example, early observational analyses comparing cancer risk among statin users and those not using statins reported reductions as large as 50–65% associated with statins; however, subsequent analyses of randomized trials, and a target trial emulation in which these design-related biases were avoided, found no effect of statins on cancer incidence⁸.

Although explicit target trial emulation can help eliminate some design-related biases, it does not eliminate data-inherent biases, including confounding due to lack of baseline randomization. In a clinical trial in which treatment is randomized, treatment groups should be comparable at baseline, but when a target trial is emulated with observational data, treatment is not randomized. As a result, treatment groups may differ at baseline in ways that affect outcomes. Problems also occur with nonadherence, when patients do not follow the treatment strategy after baseline, or with selection bias (including bias due to missing data), or bias that arises from mismeasurement. Other methods for reducing the risk of such biases must be used in addition to the target trial framework. For example, propensity score adjustment may be able to reduce the risk of bias due to confounding if appropriate data on confounding variables are available.

In a recent example, Kruger et al. emulated a target trial to estimate the comparative effectiveness of tirzepatide and semaglutide on cardiovascular outcomes for people with type 2 diabetes¹⁰. The authors emulated two completed randomized cardiovascular outcome trials to benchmark their observational analysis, after detailed protocols describing the target trial emulations were published (<https://osf.io/38rw9/overview>, NCT06659744, NCT07203677 and NCT07096063). The target trial emulations generated findings consistent with the reference trials, which increased confidence in how the interventions and outcomes were observed in US health insurance claims data. The authors then expanded their study population to lower-risk subjects treated in clinical practice and extended the

analysis to compare tirzepatide and semaglutide head to head, finding comparable effects. Since head-to-head cardiovascular outcome trials are unlikely to be conducted anytime soon, this analysis provides the best available evidence for informing clinical and policy decisions about the most appropriate drug for preventing cardiovascular events for people with type 2 diabetes.

Despite the benefits of the target trial framework, many articles do not report sufficient critical information for readers to appraise and implement the findings¹¹. The TARGET (Transparent Reporting for Studies Emulating a Target Trial) guideline provides concise instructions on how to report a target trial emulation¹². It was developed by a group of international experts in target trial emulation following the EQUATOR (Enhancing the Quality and Transparency of Health Research) recommendations. In brief, we conducted a systematic review of reporting practices in published target trial emulations¹¹, followed by a two-round survey of 18 experts in epidemiology, statistics, clinical trials and reporting-guideline development, along with policymakers and journal editors, to assess the importance of potential items for inclusion in the guideline. The group met for a 3-day consensus meeting to refine the scope of the guideline and draft the checklist, which was further refined through piloting with 108 stakeholders and potential users of the guideline, to produce the final 21-item checklist¹².

The TARGET guideline provides guidance for reporting observational studies of interventions that explicitly emulate a parallel group, individually randomized target trial. Adherence to the TARGET guideline will help readers of an observational study understand its strengths and weaknesses and will support accurate appraisal, synthesis and interpretation of study findings by decision-makers. The TARGET guideline has been recommended by the UK National Institute for Health and Care Excellence for use when reporting a target trial emulation, and journals have begun to require submission of a completed TARGET checklist, alongside submitted target trial emulations¹³. Clear reporting of observational studies of

interventions that explicitly emulate a target trial will ensure these studies can inform decisions that are unlikely to be addressed with randomized trials.

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References

1. Sherman, R. E., Davies, K. M., Robb, M. A., Hunter, N. L. & Califf, R. M. *Nat. Rev. Drug Discov.* **16**, 297–298 (2017).
2. U.S. Food and Drug Administration. FDA <https://go.nature.com/4mwELVj> (2018).
3. Hernán, M. A., Dahabreh, I. J., Dickerman, B. A. & Swanson, S. A. *Ann. Intern. Med.* **178**, 402–407 (2025).
4. Dorn, H. F. *Am. J. Public Health Nations Health* **43**, 677–683 (1953).
5. Robins, J. *Math. Model.* **7**, 1393–1512 (1986).
6. National Institute for Health Care Excellence. NICE <https://go.nature.com/41zYji7> (23 June 2022).
7. Canada's Drug Agency. CDA-AMC <https://go.nature.com/4sAagzd> (2023).
8. Dickerman, B. A., García-Albeniz, X., Logan, R. W., Denaxas, S. & Hernán, M. A. *Nat. Med.* **25**, 1601–1606 (2019).
9. Hernán, M. A. et al. *Epidemiology* **19**, 766–779 (2008).
10. Krüger, N. et al. *Nat. Med.* **32**, 342–352 (2026).
11. Hansford, H. J. et al. *JAMA Network Open* **6**, e2336023–e2336023 (2023).
12. Cashin, A. G. et al. *JAMA* **334**, 1084–1093 (2025).
13. Lumbard, H., PLOS Medicine Staff Editors. *PLoS Med.* **22**, e1004796 (2025).

Competing interests

All authors are members of the TARGET working group and developed the TARGET guideline.